Experiment : 2

Exploratory Data Analysis (EDA), Descriptive Statistics, and Data Visualization Using Python

Mapped CO(s) : CO1, CO2

Aim

To perform Exploratory Data Analysis (EDA), descriptive statistical analysis, and data visualization on the Titanic dataset using Python.

objectives

- To explore the structure of the dataset.
- To calculate descriptive statistics for numerical variables.
- To identify missing values and understand data distribution.
- To visualize the data using different graphs.
- To interpret patterns and trends from the dataset.

Prerequisites

- Basic knowledge of Python programming.
- Basic understanding of statistics.
- Python 3.x installed.
- Jupyter Notebook or Google Colab.
- Required libraries: Pandas, NumPy, Matplotlib, and Seaborn.

Theory/Background

Exploratory Data Analysis (EDA) is the process of examining a dataset to understand its structure, detect patterns, identify missing values, and discover relationships between variables before applying machine learning or statistical models.

Descriptive statistics summarize the main characteristics of data using numerical measures such as mean, median, mode, standard deviation, minimum, maximum, and quartiles. These measures help understand the distribution and variability of the dataset.

Data visualization represents data graphically, making it easier to identify trends, patterns, and outliers. Python libraries such as Matplotlib and Seaborn provide various visualization techniques including histograms, bar charts, box plots, scatter plots, and heatmaps.

EDA helps improve data quality and supports better decision-making by providing meaningful insights into the dataset.

Algorithm/Methodology

- Import the required Python libraries.

- Load the Titanic dataset into a Pandas DataFrame.
- Display the first few records and dataset information.
- Check data types and missing values.
- Generate descriptive statistics using Pandas functions.
- Visualize categorical variables using bar charts.
- Visualize numerical variables using histograms and box plots.
- Create a correlation heatmap for numerical variables.
- Interpret the observations from the visualizations.

Dataset Description

Dataset Name: Titanic Dataset

Source: Seaborn / Kaggle

The Titanic dataset contains demographic and travel information of passengers aboard the RMS Titanic. It includes passenger details such as age, gender, passenger class, fare, and survival status, making it suitable for practicing exploratory data analysis and visualization techniques.

Important Attributes

Attribute	Description
survived	Survival status (0 = No, 1 = Yes)
pclass	Passenger class
sex	Gender of passenger
age	Age of passenger
fare	Ticket fare
class	Travel class
embarked	Port of embarkation
sibsp	Number of siblings/spouses aboard
parch	Number of parents/children aboard

Program/Implementation

The program should include:

- Importing Pandas, Matplotlib, and Seaborn.
- Loading the Titanic dataset.
- Displaying dataset information.

- Checking missing values.
- Calculating descriptive statistics using describe().
- Creating histograms, bar charts, box plots, and scatter plots.
- Generating a correlation heatmap.
- Interpreting the visualizations.

Test Cases

Test Case	**Expected Result**
Load Titanic dataset	Dataset loads successfully
Display dataset information	Data types and missing values displayed
Generate descriptive statistics	Mean, median, standard deviation, etc. displayed
Plot histogram	Distribution of numerical variable displayed
Plot box plot	Outliers identified
Generate heatmap	Correlation between numerical variables displayed

Experimental Results

Observation

- The Titanic dataset was successfully loaded and explored.
- Descriptive statistics summarized the numerical features of the dataset.
- Missing values were identified in some columns.
- Histograms showed the distribution of Age and Fare.
- Box plots helped identify outliers.
- Heatmaps displayed the correlation among numerical variables.
- Data visualizations made it easier to understand passenger characteristics and survival patterns.

Result

Exploratory Data Analysis, descriptive statistical analysis, and data visualization were successfully performed using Python. The analysis provided useful insights into the Titanic dataset and demonstrated the importance of visualizing data before further analysis.

Conclusion

The experiment demonstrated the use of Python libraries for performing Exploratory Data Analysis (EDA), descriptive statistics, and data visualization. Various statistical measures and graphical techniques helped understand the structure, distribution, and relationships within the Titanic

dataset. EDA serves as an essential step in the data science process by improving data understanding and supporting informed decision-making.

Exercises/Assignments

1. Calculate descriptive statistics for the **Fare** column.
2. Create a histogram of the **Age** variable.
3. Draw a box plot for **Fare** to identify outliers.
4. Generate a heatmap showing the correlation between numerical variables.
5. Interpret any one visualization created during the experiment.